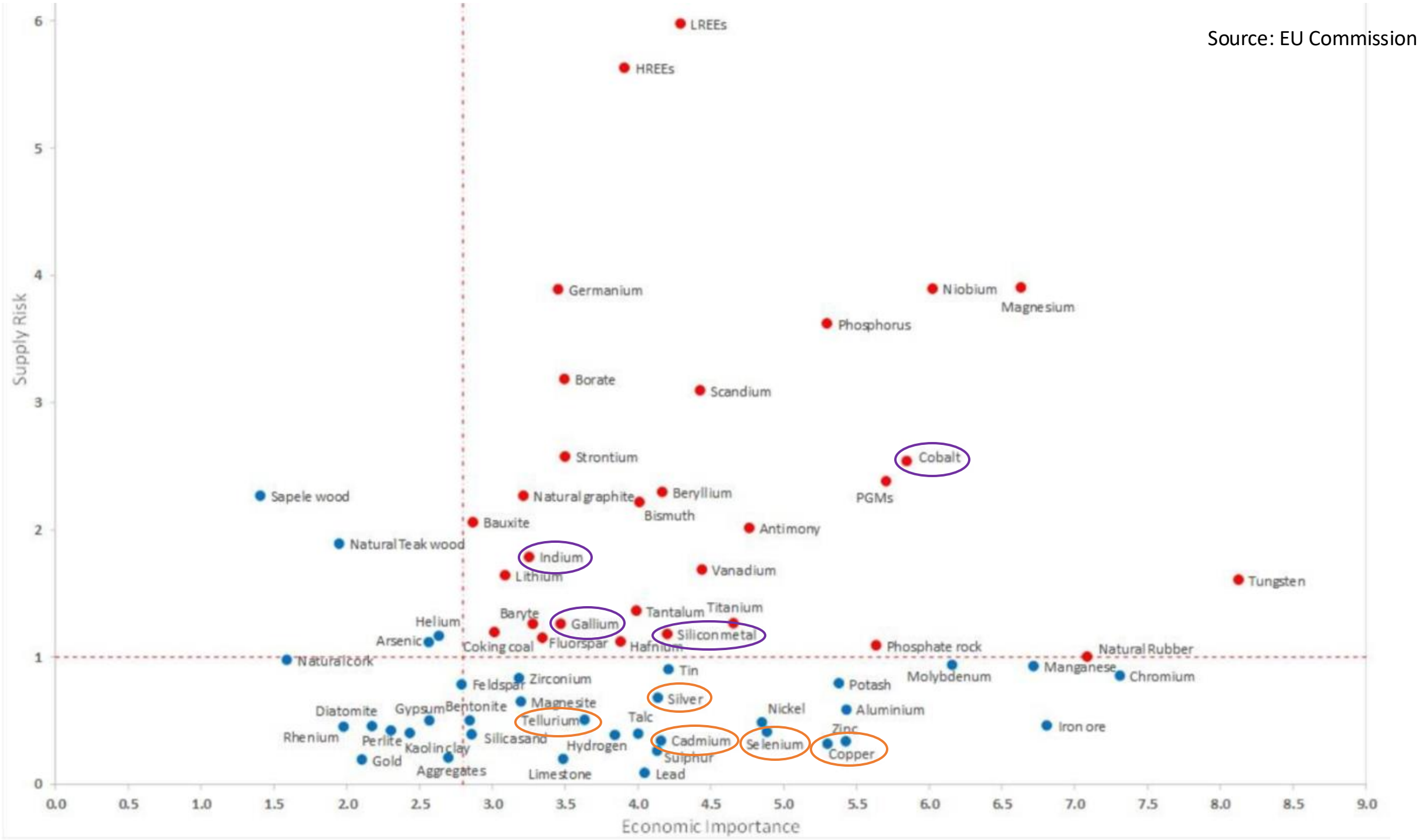
What elements are important in solar (PV) technology?

Si (glass)
Cd (films)
Te (films)
Cu (films, interconnectors)
In (films)
Ga (films)
Se (films)
Ag (films)

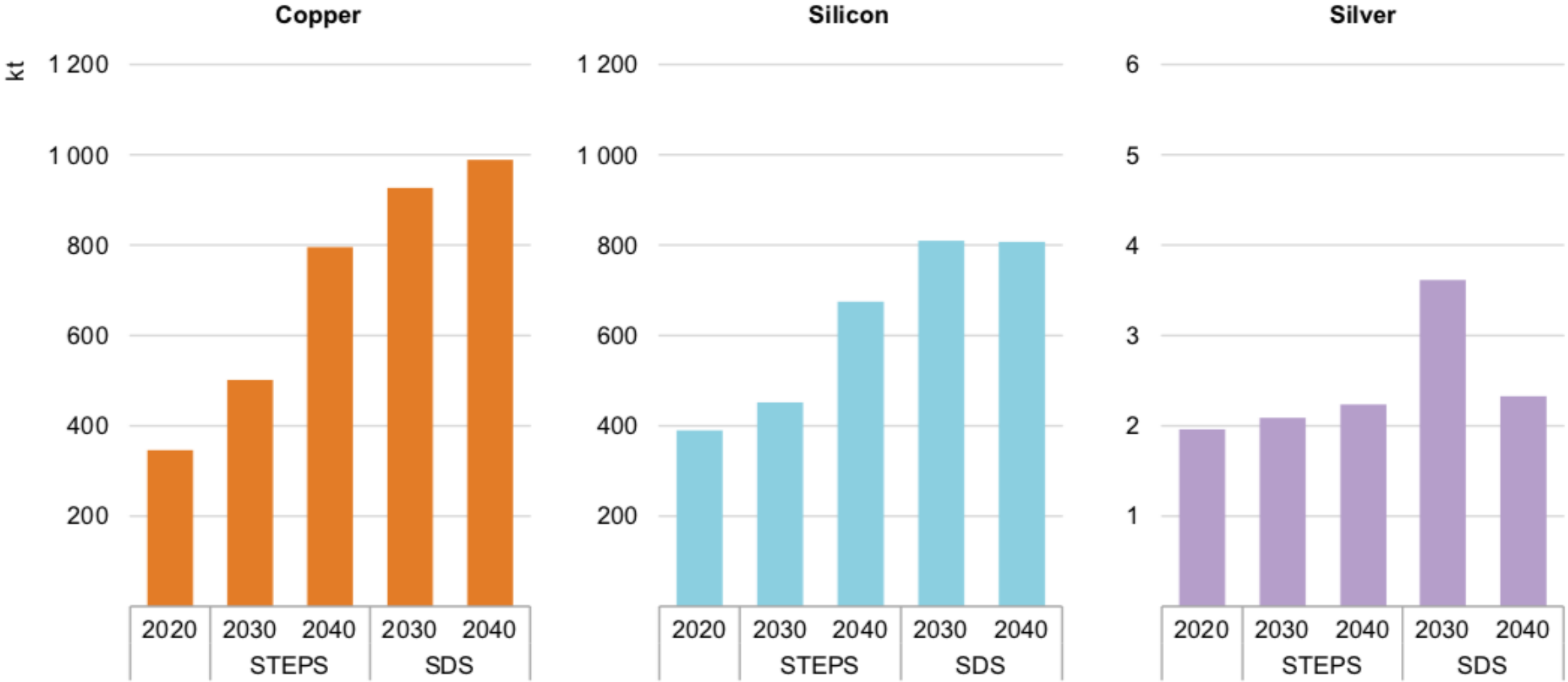
Si cells dominate PV market
Si and Ag are among most expensive in PV cells => future advances in material intensity to reduce amounts needed
Perovskite ($CaTiO_3$) solar cells may be alternative, but have technical drawbacks – soluble in water = unstable, highest efficiency at small size, so need to be scaled up

Source: EU Commission



Solar PV: Copper demand more than doubles by 2040, but continued innovation to reduce mineral intensity helps to offset demand growth for silicon and silver

Demand for copper, silicon and silver for solar PV by scenario



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Note: kt = thousand tonnes.

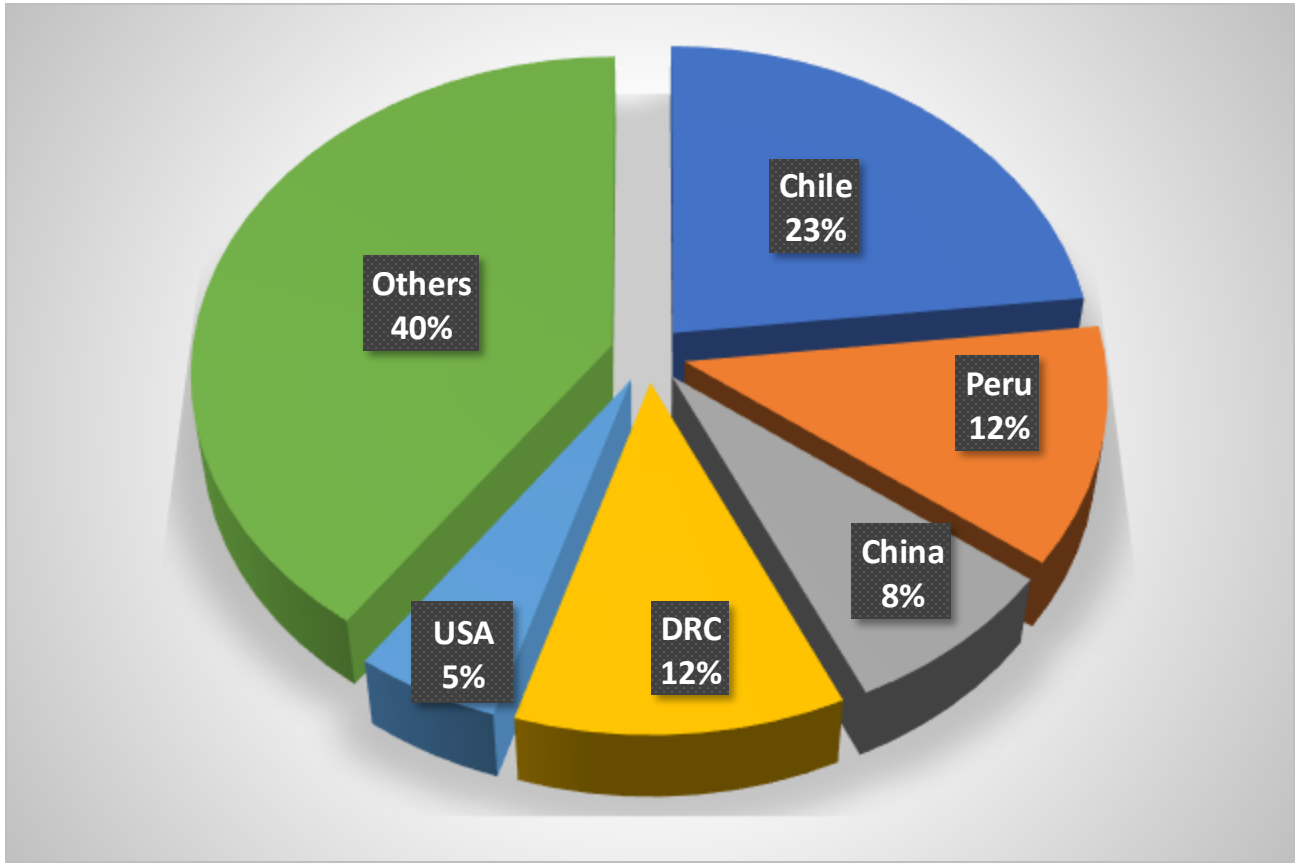
Cu

- Most widely used in energy transition
- unmatched thermal and electrical conductivity
- broad range of electronic and industrial applications. Its attributes make it challenging to substitute, although substitutions exist by Al (electrical equipment, power cables), optical Fiber (telecoms), Ti (heat exchangers)

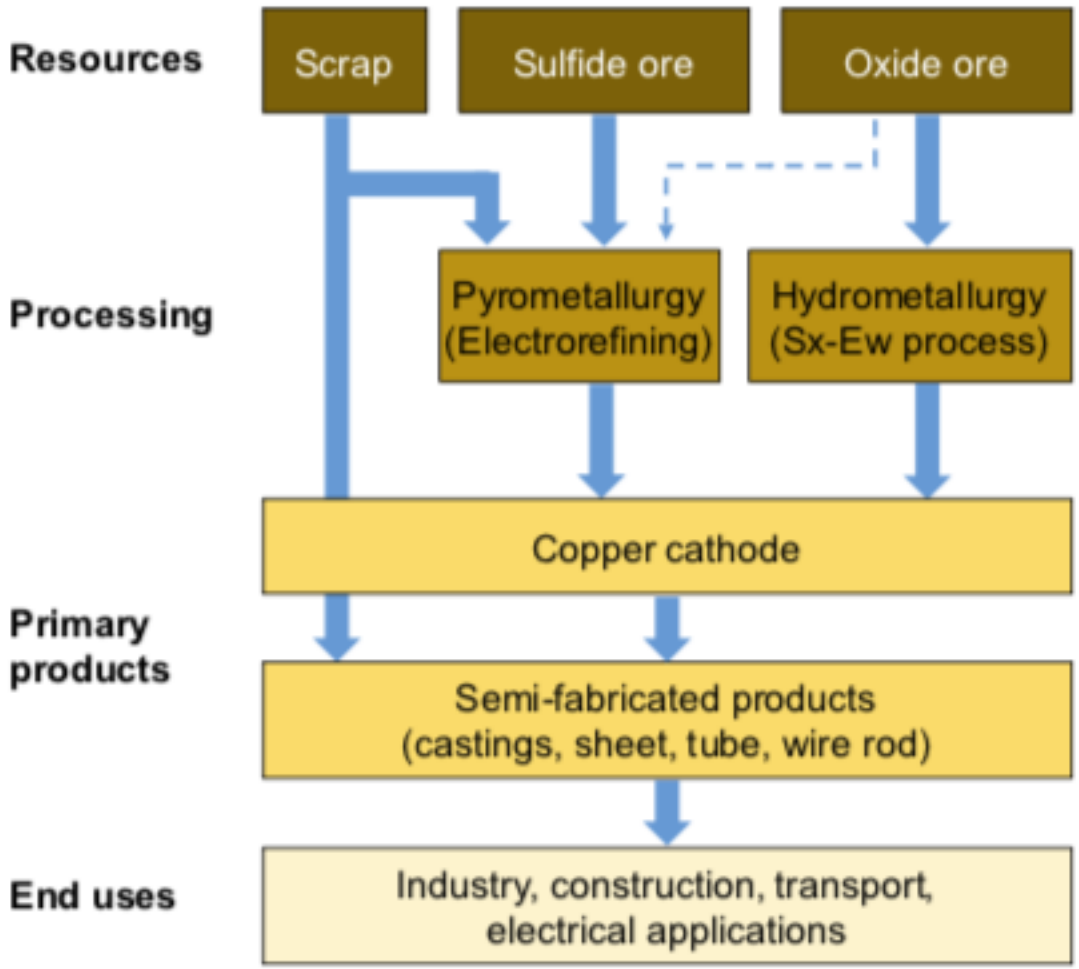
Cu

- 30% increase in production over last 10 years to match demand
- New projects in pipeline to 2025, but beyond uncertain
- Some big mines like Escondida expected to see peak production soon
- Mined Cu grades decreased from 10-20% in 19th century to <1% now; more research and technology have unlocked reserves, and processing innovations have allowed lower grade processing
- High water demand for processing, but water is scarce in many producing countries
- Arsenic contamination at lower grades and pollution

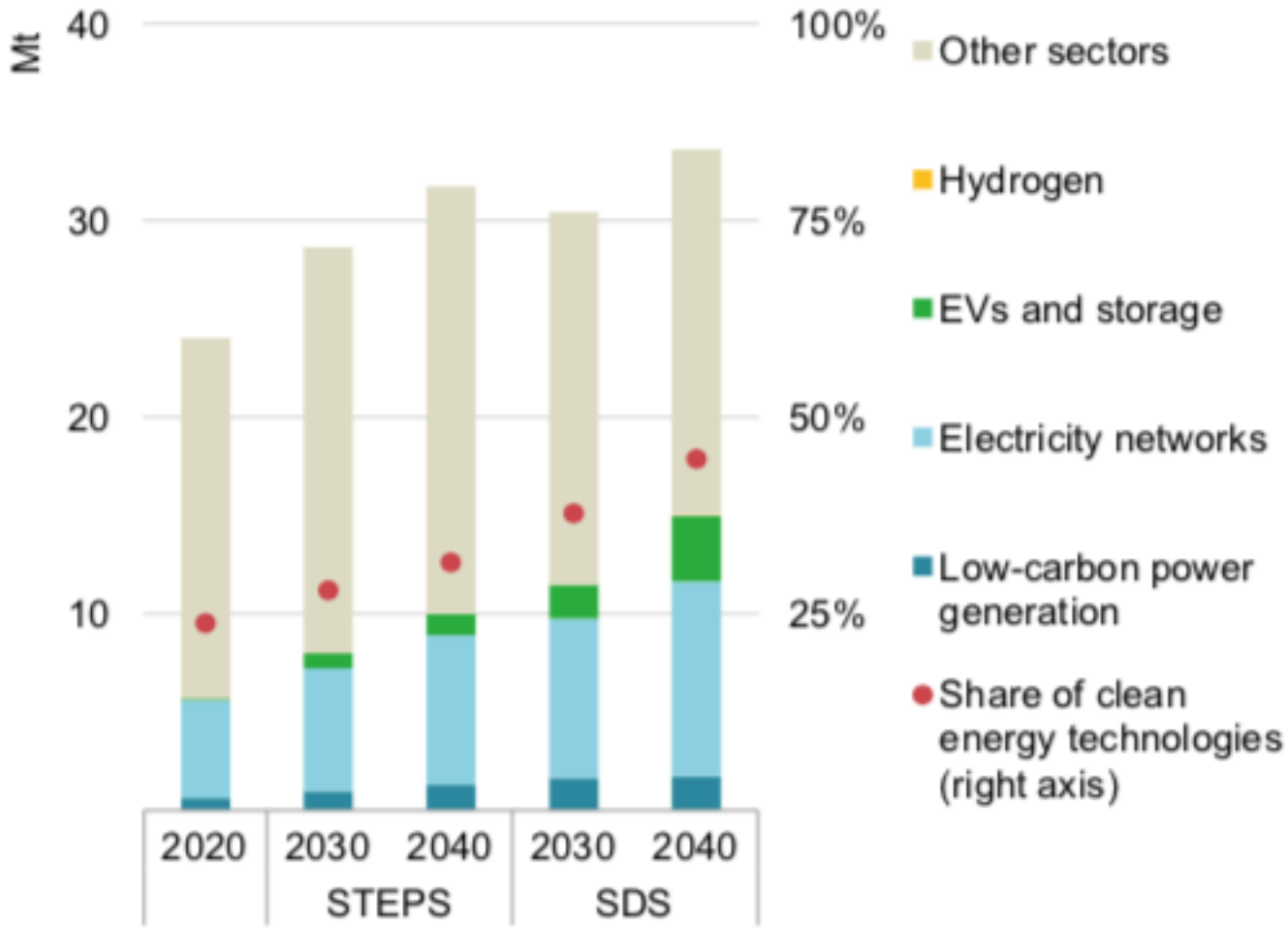
Cu production 2023 (USGS)



Copper supply chain



Total copper demand by sector and scenario



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Notes: EVs = electric vehicles. Sx-Ew = solvent extraction and electrowinning. High-grade oxide ore is processed in pyrometallurgy. Demand does not include the volume reused in a semi-fabricated form.

Cu challenges

- Challenging to substitute due to its superior performance in electrical applications
- Mines currently in operation are nearing their peak due to declining ore quality (decreases grades, low grade ores economic?) and reserves exhaustion (move to fringes of deposits)
- Declining ore quality exerts upward pressure on production costs, emissions and waste volumes (processing on site, dust suppression, deeper mining = higher costs)
- Mines in South America and Australia are exposed to high levels of climate and water stress

Critical because:

- Supply
- (projected) demand in energy transition
- Low substitutability

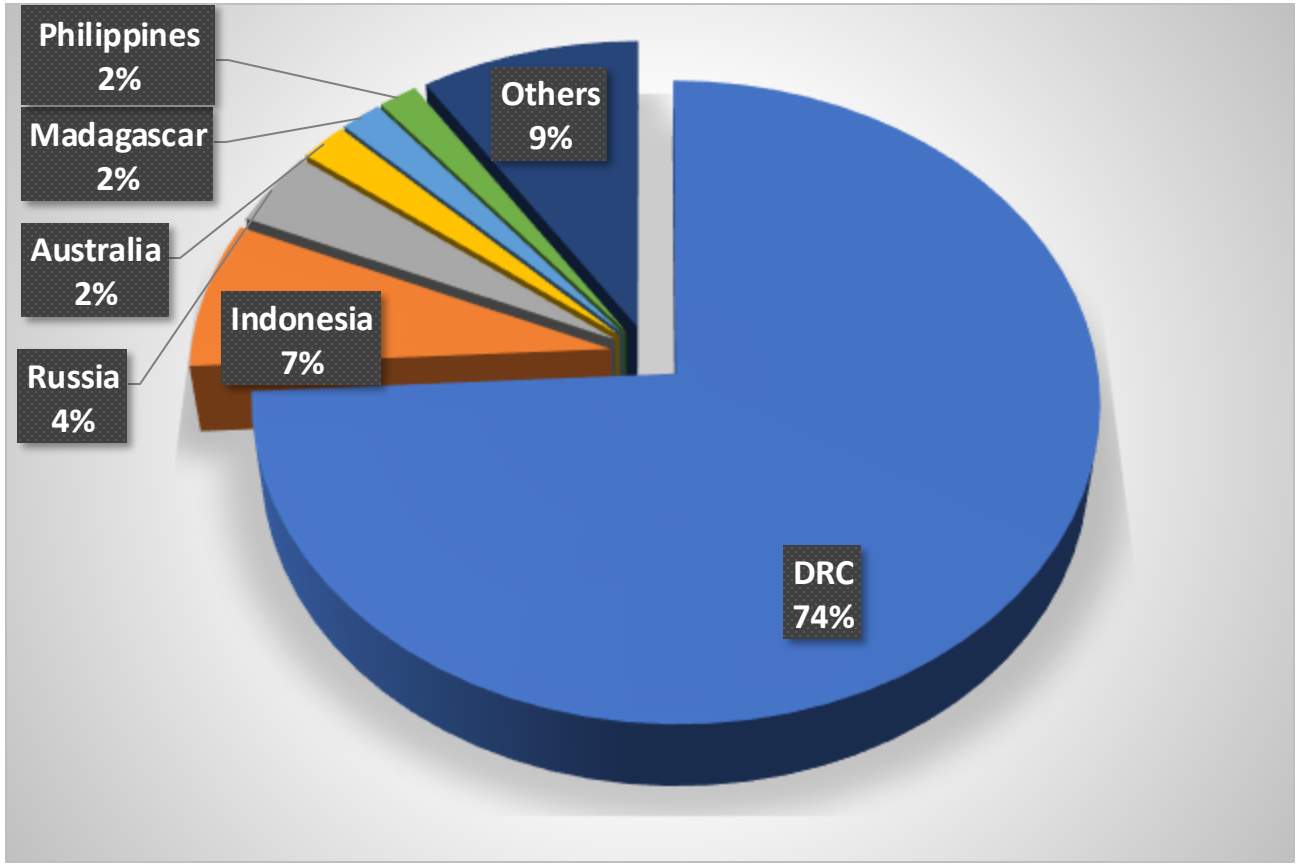
Co

- Use in Li-ion batteries for EV, also alloys, carbide, magnets
- unmatched thermal and electrical conductivity
- broad range of electronic and industrial applications. Its attributes make it challenging to substitute

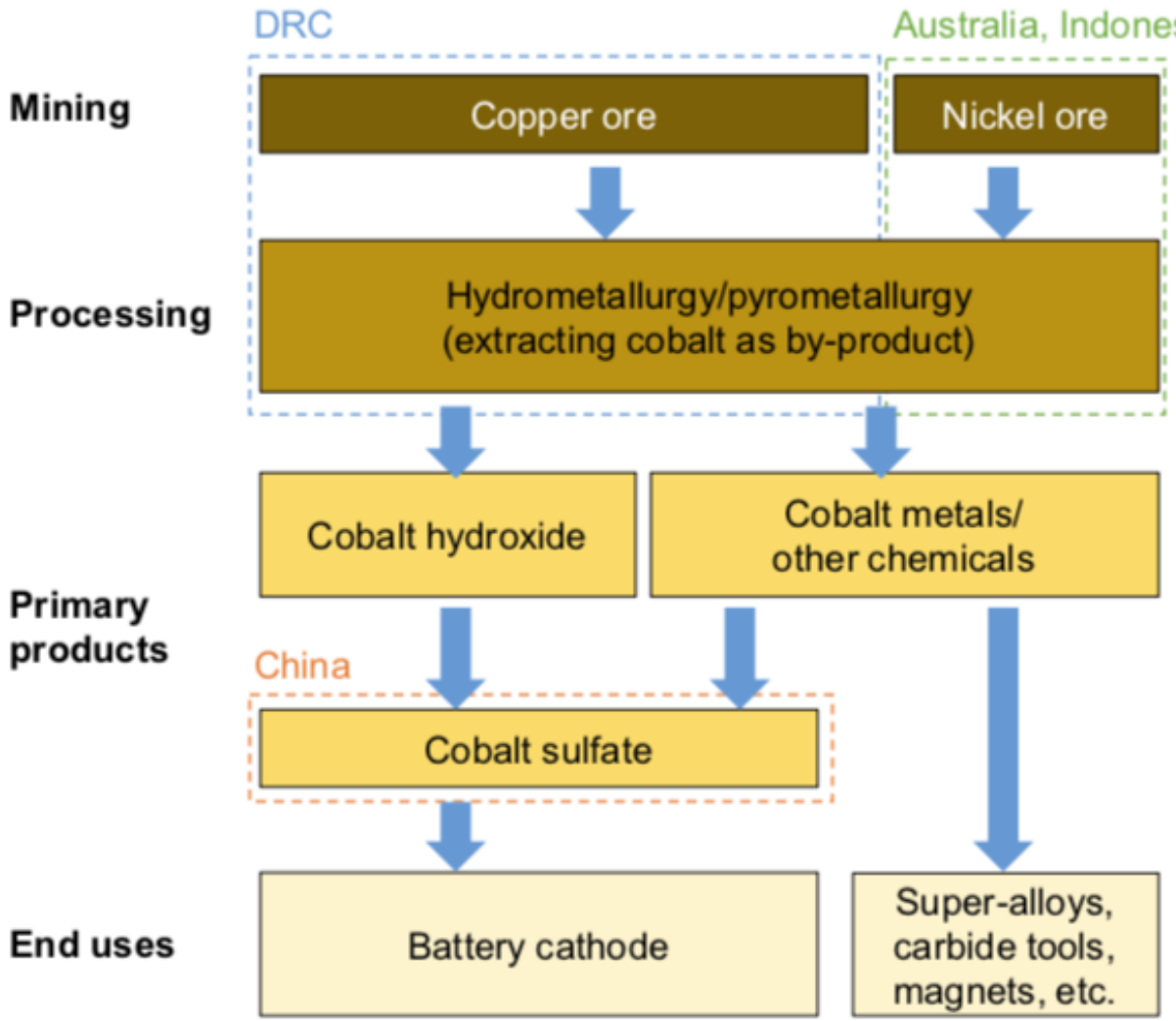
Co

- Unexpected demand rattled market -> strain on supply
- Production increase 10% in last 5 years
- 74% DRC, by-product of Cu/Ni mines
- 10-20% of DRC mining from artisanal mining
- China main processing country
- Demand depends on cathode technology in batteries (could turn to Ni)
- Expected output from current mines and projects likely to satisfy demand (STEPS, not SDS scenario)
- DRC and China are linked via Co – supply chain issues, China has invested in DRC

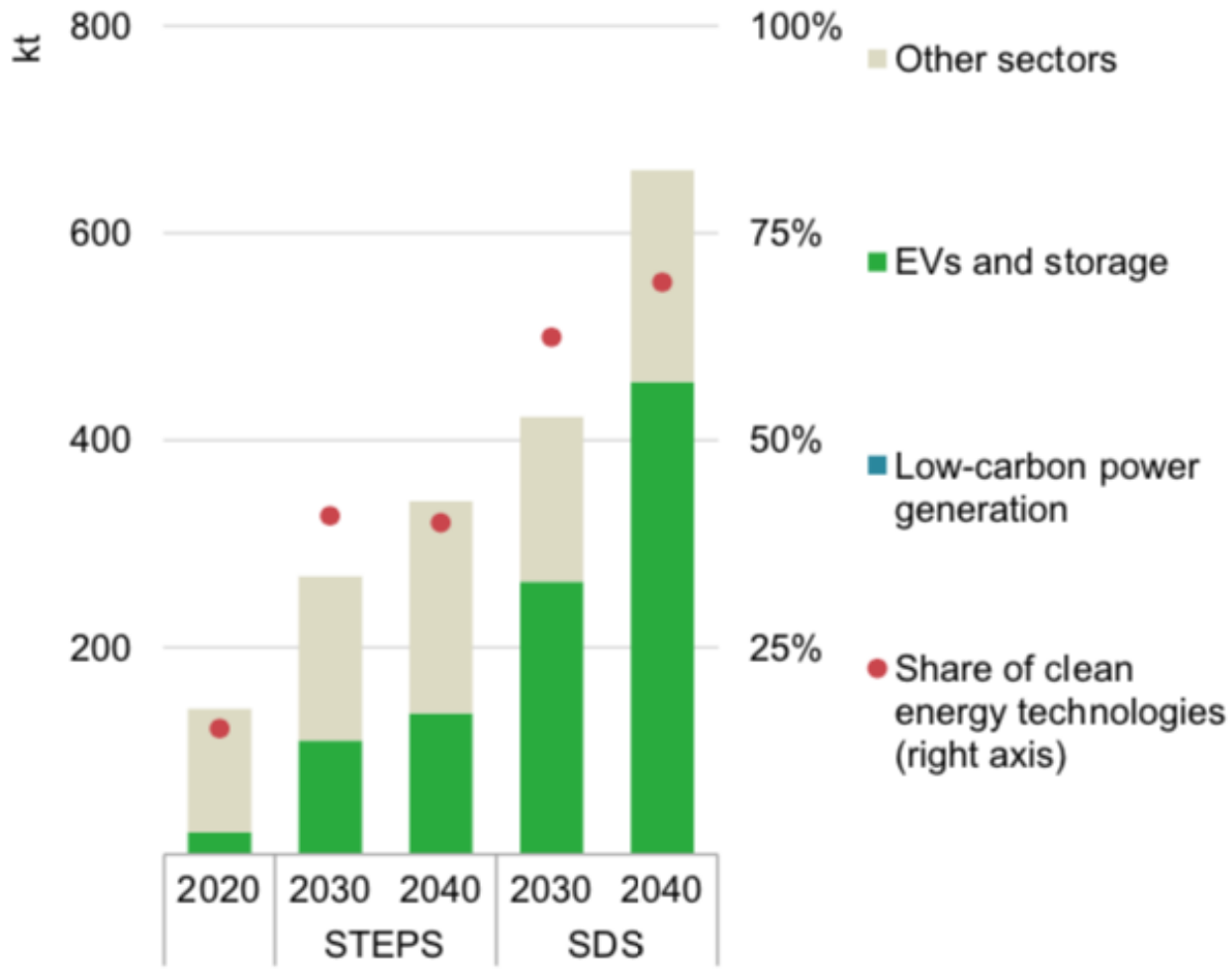
Co production 2023 (USGS)



Cobalt supply chain



Total cobalt demand by sector and scenario



Artisanal mining (ASM)

- Positive effect on markets as production can shift quickly
- Negative effect on market, as vulnerable to economic and social events
- Unregulated, unsafe working, child labor, intermediate traders keep prices low and reap profits
- Yet provides income locally – can mitigate root cause of child labor, which is poverty, ideally organized locally, contributing to local development
- Efforts to formalize work via governments, consumer organizations, industry efforts
- Industry has normally secured best licenses; ASM will not happen in remaining licenses with low grade, but tend to occur illegally on industry licenses





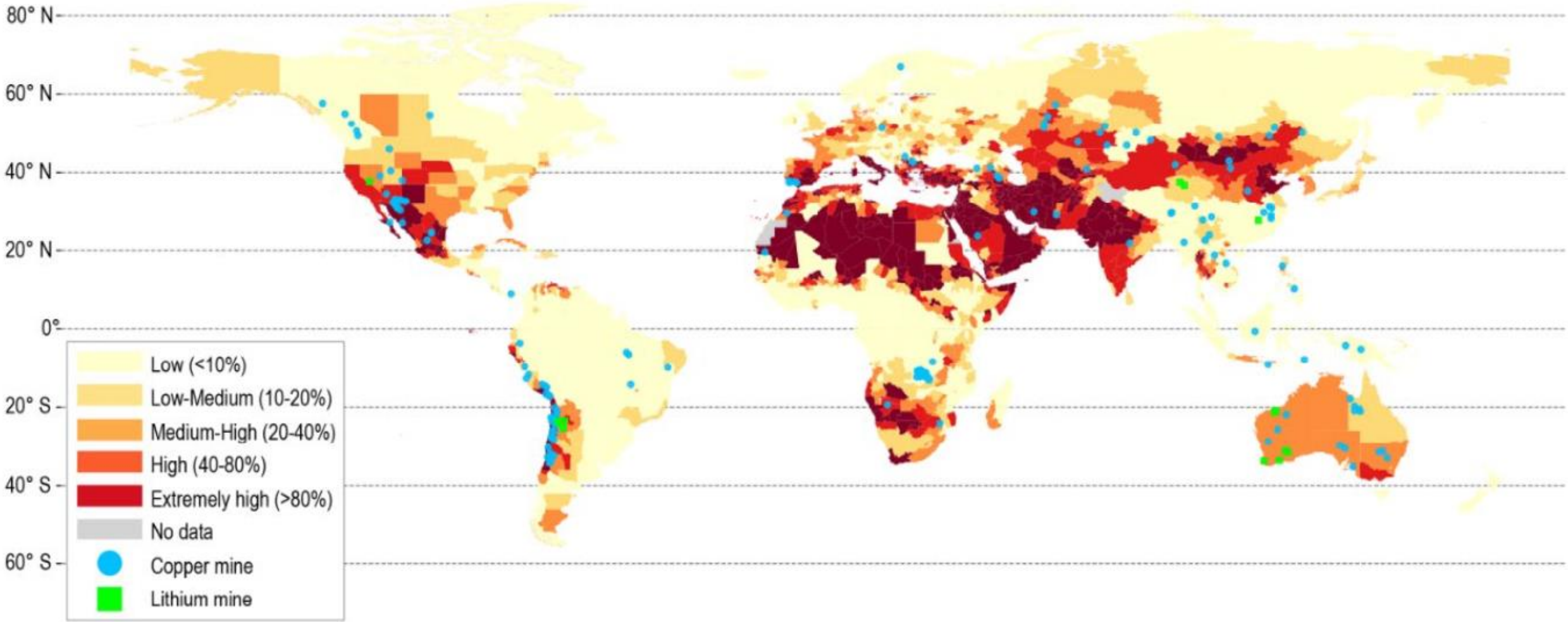
Co challenges

- High reliance on DRC for production and on China for refining set to persist, as only a few projects are under development outside these countries
- Significance of artisanal small-scale mining makes the supply vulnerable to social pressures
- New supply is subject to developments in nickel and copper markets as some 90% of cobalt is produced as a by-product of these minerals

Critical because:

- Supply
- (projected) demand in energy transition
- By-product
- Social/environmental pressures

Location of copper and lithium mines and water stress levels, 2020



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